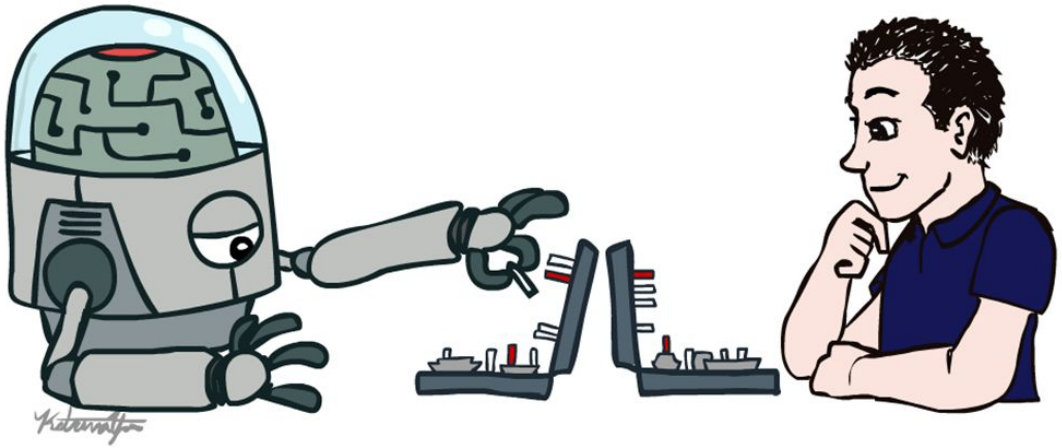
Artificial Intelligence

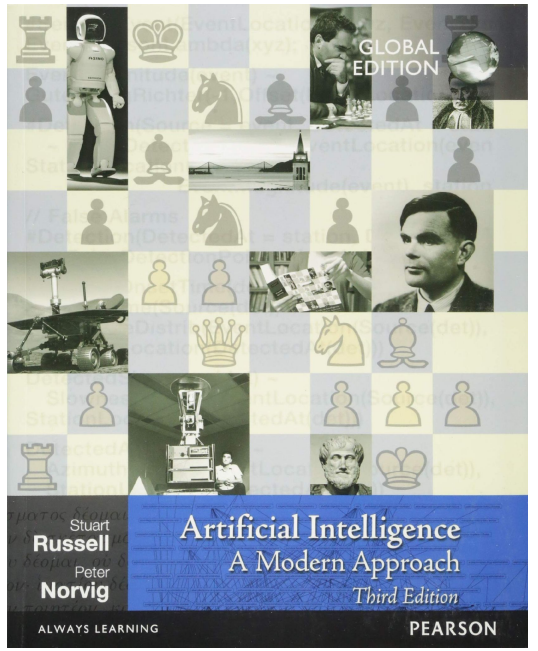
Introduction



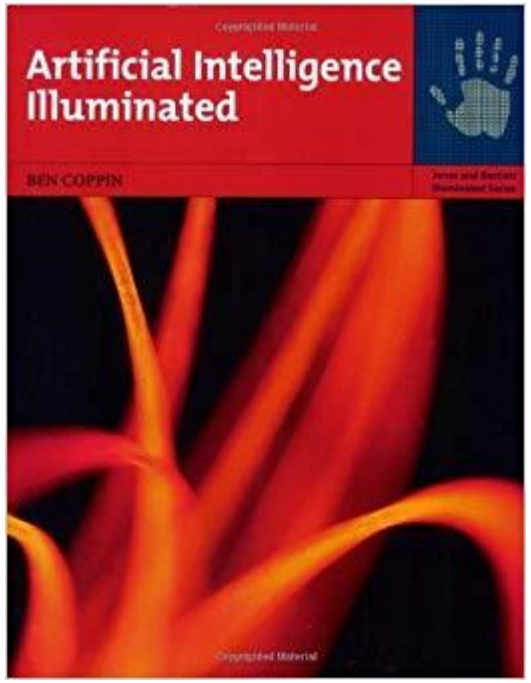
Modified slides of CS-188 Berkeley University

Recommended Books

- **Artificial Intelligence: A Modern Approach**,
  Russell & Norvig.



- "**Artificial Intelligence Illuminated**"
  Ben Coppin,
  (http://futuresoft.yolasite.com/resources/Artificial%20Intelligence%20Illuminated.pdf)



Recommended Books

- “Artificial Intelligence – Structures and Strategies for Complex problem solving”, George F. Luger, Pearson International Edition, Sixth edition, 2009.
- “Artificial Intelligence: A new synthesis” Nils Nilsson, Morgan Kaufmann, 1998
- **UC Berkeley CS188 Intro to AI** (http://ai.berkeley.edu/home.html)

Grading (Tentative)

ITEM	WEIGHTAGE
Quizzes	15 %
Assignments	15 %
Class Participation	5%
Mid Term	20 %
Final Term	45 %
Total	**100**

Logistics

- Quiz may be un-announced.
- Read the book
- Plagiarism – Write your own Solution
- Deadline of the assignments will not be extended
- No Late submissions will be accepted.

Other Announcements

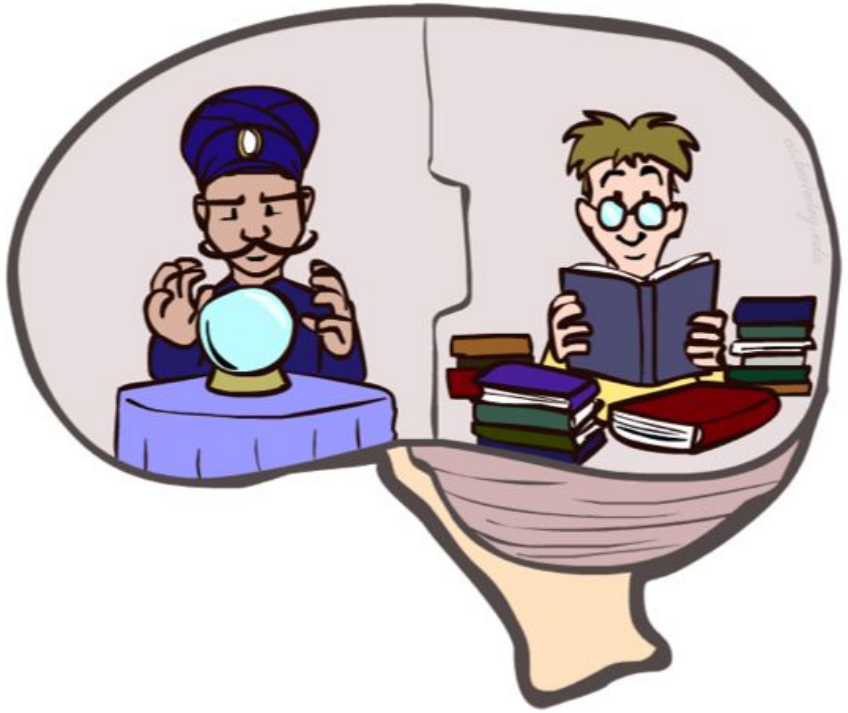
- Email address:
  - maham.armaghan@ucp.edu.pk
- Office:
  - D Block 2nd Floor
- Course Coordinator
  - Dr. Mohsin Ashraf

Course Topics

- Part I: Solving Problems by Searching
  - Fast search / planning
  - Constraint satisfaction
  - Heuristics
  - Adversarial and uncertain search

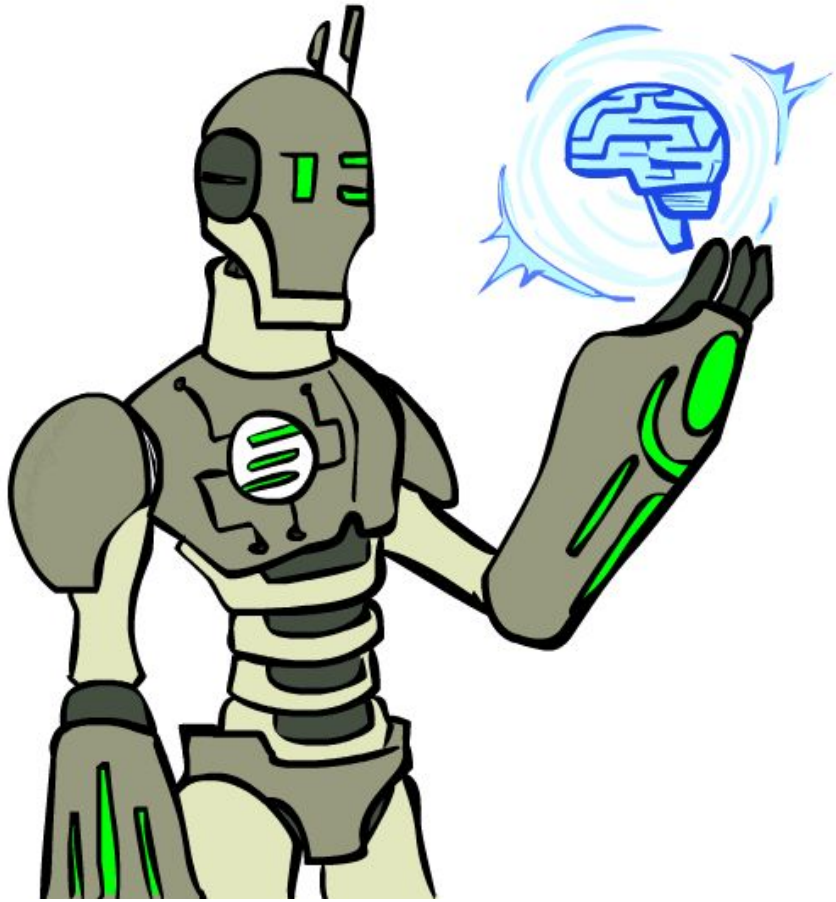
- Part II: Knowledge, reasoning, and Planning
  - Reasoning agents
  - Propositional logic
  - Predicate logic
  - Knowledge-based systems

- Part III: Machine Learning
  - Classification
  - Clustering
  - Neural Networks



Overview

- What is artificial intelligence?
- What can AI do?
- What is this course?



Question

- Is this mouse intelligent?
  - It depends on context
  - Route planning (Yes)
  - Mathematical problem solving (No)



- Is Google Intelligent?
  - Can google drive a car?
  - Task specific intelligence

- Are humans intelligent?

WHY AI?

- Why can't we simply program every possible situation?
- Imagine making software for
  - **Self-driving cars**
  - You can write
    - IF person crosses road
    - THEN stop
  - But what about
    - fog
    - rain
    - child running
    - broken traffic light
    - cyclist
    - ambulance
    - dog
  - Thousands of possibilities. Traditional programming starts failing.
- AI helps machines
  - learn
  - reason
  - decide

Why AI?

- **Two main goals of AI:**
  - To create useful “smart” programs able to do tasks that would normally require a human expert
  - To understand human intelligence better as we test theories of human intelligence by writing programs which emulate them

What is Intelligence?

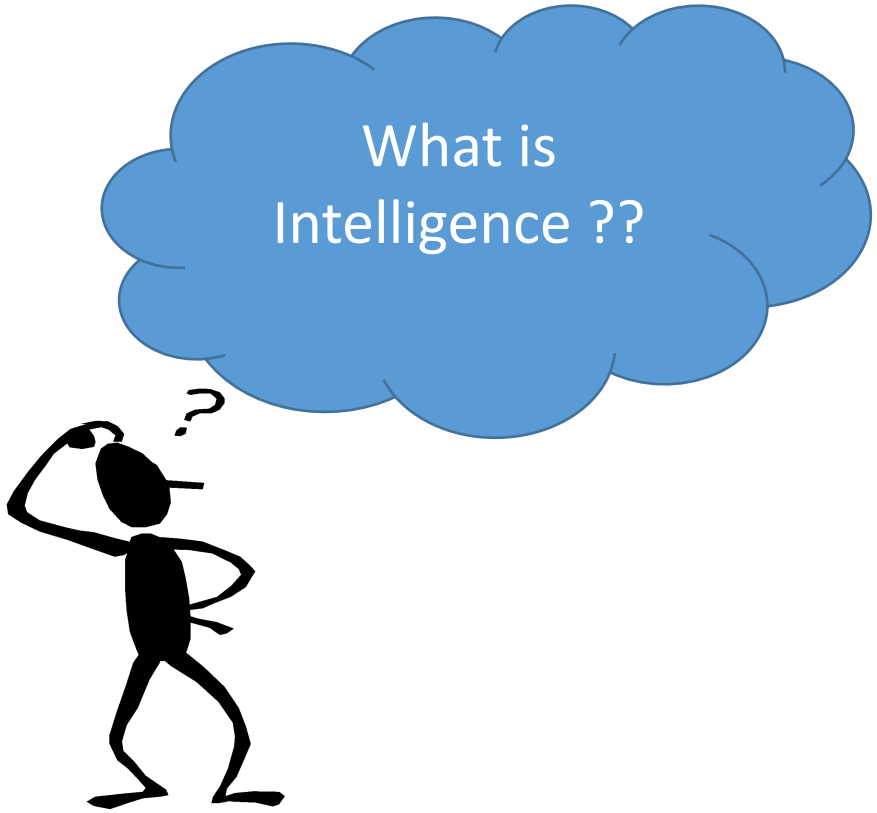
- Ability of problem solving
  - Example: Finding the shortest path to university.
- Ability to think, plan and schedule
  - Preparing timetable before exams.
- Efficient memory and information manipulation
  - Remembering previous mistakes.
- Ability to tackle ambiguous and fuzzy problems
  - A fuzzy problem does not have crisp, exact values — it deals with “how much” rather than “yes or no.”
- Ability to understand and perceive

Artificial Intelligence

- "***Study of computations that make it possible to perceive, reason and act***."
- Branch of *Science* which deals with **helping machines to find solutions to complex problems in a more human-like fashion**

- In short, putting human intelligence into machines

Artificial Intelligence



- Perceive
- Understand
- Remember
- Think
- Reason
- Learn from Experience
- Adapt to the new Situation
- Problem Solving

Humans Vs Machines

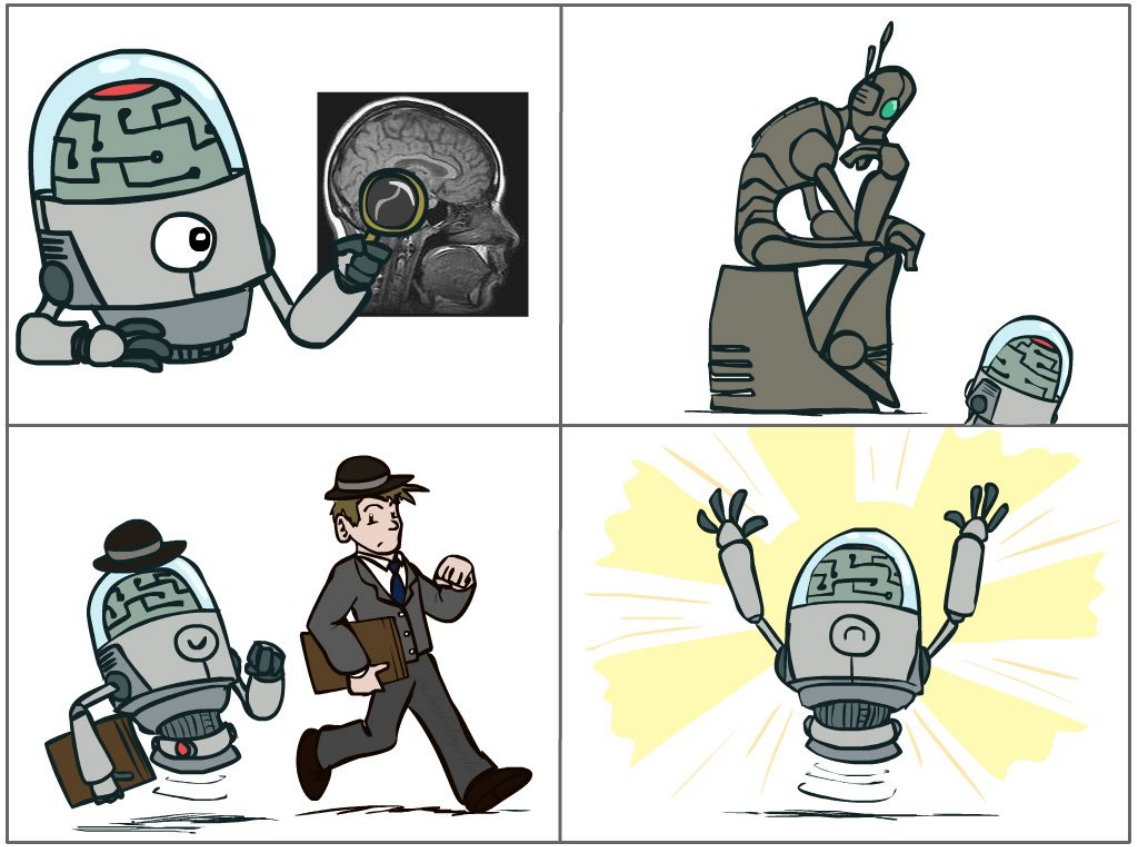
Humans	Machines
• Symbolic calculation • Natural language understanding • Not very precise • Knowledge • Generalize from examples • Deal with noisy inputs	• Numeric calculation • Machine Language • Precise • Data • Cannot generalize • Cannot deal with noise

What is AI?

The science of making machines that:

Think like people

Cognitive modeling approach



Think rationally

Laws of thought approach

Act like people

Turing Test approach

Act rationally

Rational agent approach

What is AI?

- Cognitive Modelling Approach:
  - This approach tries to answer: “how do humans think?” and then replicate the thinking process. Based on Psychology, neuroscience, and cognitive science.
- Law of thought Approach:
  - Use mathematical logic to derive correct conclusions. (Propositional Logic, Predicate Logic, Theorem)
  - Example:

    All humans are mortal
    Socrates is human
    → Socrates is mortal
- Turing Test:
  - If a machine behaves like a human, it is intelligent.

  Limitation:
  - It measures imitation, not true intelligence.
  - A system can trick humans without real understanding.
- Rational Agent Approach:
  - Acts to maximize expected performance given what it knows.
  - Example:
    - Self-driving car:
    - Doesn’t need to drive like a human
    - It just needs to drive safely and efficiently

What is Artificial Intelligence

Thinking Humanly	**Thinking Rationally**
"The exciting new effort to make computers think . . . *machines with minds*, in the full and literal sense." (Haugeland, 1985) "[The automation of] activities that we associate with human thinking, activities such as decision-making, problem solving, learning . . ." (Bellman, 1978)	"The study of mental faculties through the use of computational models." (Charniak and McDermott, 1985) "The study of the computations that make it possible to perceive, reason, and act." (Winston, 1992)
Acting Humanly	**Acting Rationally**
"The art of creating machines that perform functions that require intelligence when performed by people." (Kurzweil, 1990) "The study of how to make computers do things at which, at the moment, people are better." (Rich and Knight, 1991)	"Computational Intelligence is the study of the design of intelligent agents." (Poole *et al.*, 1998) "AI . . . is concerned with intelligent behavior in artifacts." (Nilsson, 1998)

Figure 1.1 Some definitions of artificial intelligence, organized into four categories.

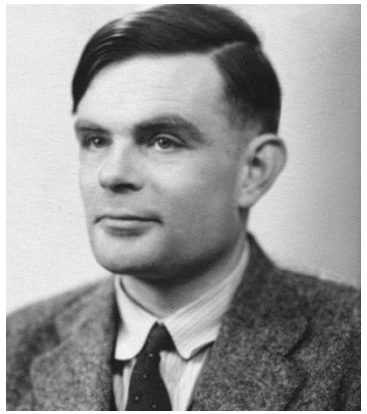
What is Artificial Intelligence?

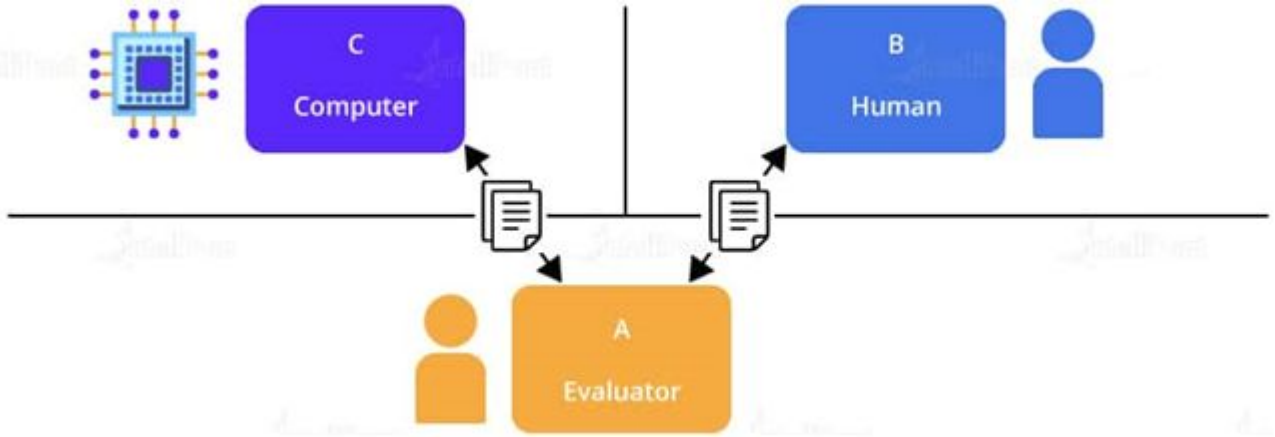
- Artificial intelligence (AI) refers to the development of computer systems that can perform tasks that typically require human intelligence. These tasks include understanding natural language, recognizing patterns, learning from experience, reasoning, and making decisions

Acting Humanly: Turing Test

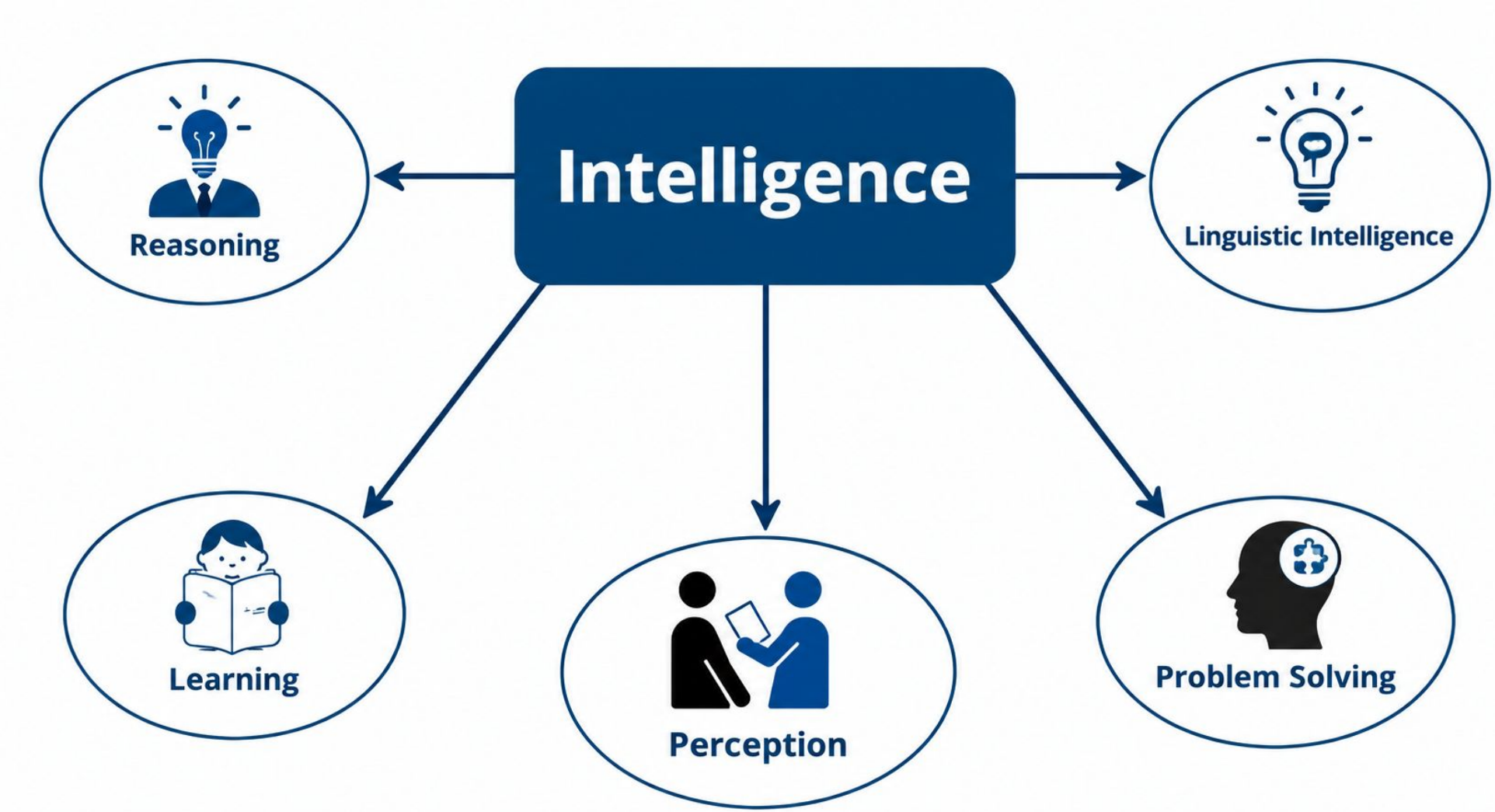
- Alan Turing (1950) "Computing machinery and intelligence".

A computer passes the test if a human interrogator, after posing some written questions, cannot tell whether the written responses come from a person or from a computer.





Turing Test Capabilities



1

TURING TEST CAPABILITIES

Linguistic Intelligence

- **Understand human language**
  The machine must understand the meaning of words, sentences and context.
- **Generate meaningful responses**
  It should produce answers that are relevant, coherent and human-like.
- **Handle grammar, context, tone**
  It must use correct grammar and adapt its tone based on the situation.



Example:
If the judge asks: "Why do people feel nervous before exams?"
The machine must understand the context and provide a human-like explanation.

Reasoning

The machine must use logic to connect facts, draw conclusions and answer analytical questions.

- **Draw logical conclusions**
- **Answer analytical questions**
- **Connect facts**



Example:
If asked "All cats are mammals. Milo is a cat." Is Milo a mammal?"
The system must reason logically.
Reasoning allows intelligent responses — not random replies.

2

TURING TEST CAPABILITIES

Learning

To appear human, a system must:

- Adapt to new questions
- Improve with experience
- Update its knowledge

If the judge changes topic suddenly, the machine must adjust.



Perception

In the basic Turing Test, pecreption isn't required (text-only). But in the Total Turing Test, the machine must also:

- Recognize images
- Understand speech
- Interpret visual information
- If shown a picture and asked:

"What is happening here?"
The system must interpret visual input.



Problem Solving

The system must:

- Solve puzzles
- Answer reason-based questions
- Handle abstract thinking

Example: If today is Monday, what will it be in 10 days?



★ These capabilities together help a machine behave intelligently and human-like in different situations.

Thinking Humanly: The cognitive modeling approach



- Requires scientific theories of internal activities of the brain.
  - Cognitive Science: Predicting and testing behavior of human subjects (Hypothesis based)
  - Cognitive Neuroscience: Direct identification from neurological data (Data driven)
- However:
  - the available theories do not explain anything resembling human-level general intelligence – can not understand full human creativity, general intelligence, flexible reasoning
  - It may be impossible to identify the detailed structure of human problem solving using only externally-available data. We can not fully know exact mental steps, all neural connections, complete internal representation because thoughts are not directly observable, brain is extremely complex.

Thinking Humanly:
The cognitive modeling approach

- For that we must determine how humans think.
- Need to get *inside* the actual workings of human minds.

Three ways to do this:

- Through introspection
  - trying to catch our own thoughts as they go by
    - When solving 27 × 14, what happened in your mind?
- Through psychological experiments
  - observing a person in action (color-word)
- Through brain imaging
  - observing the brain in action.



Thinking Rationally: Laws of Thought



- Aristotle: First to codify "right thinking"
  - Right thinking: Thinking according to correct logical rules
- Several Greek schools developed various forms of logic:
  - Notation and rules of derivation for thoughts
  - Propositional logic, Predicate logic, etc.
- Always yield correct conclusions when given correct premises, logically wrong premises can lead to wrong results
  - E.g. "Socrates is a man; all men are mortal; therefore, Socrates is mortal."
- These laws of thought were supposed to govern the operation of the mind; their study initiated the field called **logic**.

Thinking Rationally: Laws of Thought

- **Problems:**
- Not easy to state informal knowledge in logical notation
  - How do you translate "The soup is a bit too salty" or "She seems upset" into precise logical notation? How would you write these statements in mathematical logic?
  - There is no clear answer because words like:
    - Bit
    - Seems
    - Quite
    - Almost
    - Probably

  are subjective
- Classical logic works with statements that are either:
- True
- False
- Is 30°C hot?
  - Depends on
    - Weather
    - Person
    - Country
- Humans understand this naturally.
- Computers struggle.

Thinking Rationally: Laws of Thought

- Mostly work for toy examples
  - A simplified, small, artificial problem created for teaching or testing ideas — not a real-world complex situation.
- Big difference between solving a problem "in principle" and solving it “in practice”
  - Problems with just a few hundred facts can exhaust the computational resources of any computer

Thinking Rationally: Laws of Thought

- How to measure if a program is thinking rationally
- Can we see someone's thoughts?
  - Student A solves a math problem using one method.
  - Student B solves it using another method.
  - Both get the correct answer.
  - Can we say whose thinking was better?
  - No. We judge them based on the **result**, not their internal thought process. The same applies to AI.
- We evaluate:
- Did the robot reach the destination?
- Did the chess program win?
- Did the medical AI diagnose correctly?

Conclusion:

Right Thinking
Must have rules and logic
Complex and Hard

Acting rationally: Rational agent

- An **agent** is anything that perceives its environment through sensors and acts upon that environment through actuators.
  - Actuators: The mechanism through which an agent acts upon its environment.
- Sensors □ Agent □ Actuators
  - Sensors = Perception (input)
  - Actuators = Action (output)
- **Self-Driving Car**
- Sensors:
  - Camera
  - Radar
  - Lidar
- Actuators:
  - Steering wheel control
  - Brakes
  - Accelerator
- The car **perceives** the road using sensors and **acts** by moving steering/brakes via actuators.

Acting rationally: Rational agent

All computer programs do something, but computer **agents** are expected to do more:

1. Operate autonomously (No constant human control)

   Example:

   A calculator waits for input. (Not an agent)

   A thermostat automatically adjusts temperature.

2. Perceive their environment

   Example:

   A robot vacuum:

   Environment → Floor, walls, furniture
   Sensors → Infrared sensors, bump sensors
   Perception → Detects obstacle
   Action → Changes direction

   If it cannot sense obstacles, it cannot behave intelligently.

3. Persist over a prolonged time period

   Program → Runs → Stops
   Agent → Continuous operation

   Example:

   A sorting program runs once and finishes. (Not an agent)

   A spam filter keeps running daily.

Acting rationally: Rational agent

4. Adapt to change, and create and pursue goals.

- Environment changes → Agent updates behavior
- Goal → Plan → Action → Evaluate → Adjust

Example:

If traffic increases, navigation app chooses new route.

If opponent changes strategy, game AI adapts

Example:

Ordinary Program: Follow fixed instructions

Agent: Choose actions to achieve a goal

5. Make correct inferences

Example:

If road is wet → infer it might be slippery.

If user searches “cold medicine” → infer user may be sick.

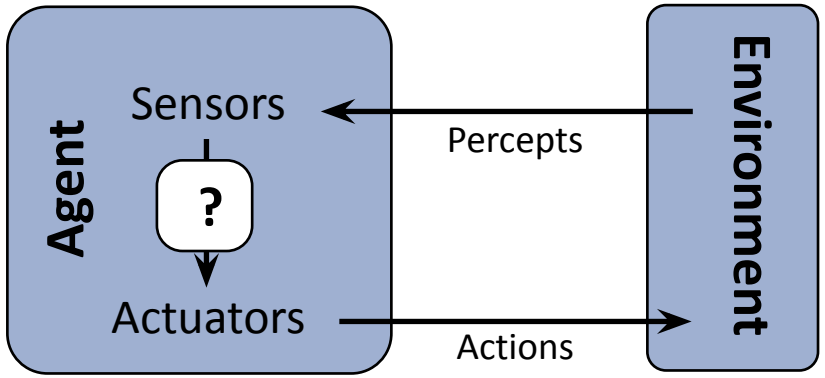
Acting rationally: Rational agent

- A **rational agent** is one that acts so as **to achieve the best outcome** or, when there is uncertainty, the **best expected outcome**.
- Rational behavior: **doing the right thing**
- The right thing: the optimal (best) thing that is expected to maximize the chances of achieving a set of goals, in a given situation
- Making correct inferences is part of being a rational agent
- There are also ways of acting rationally that cannot be said to involve inference -- Reflex actions.

Acting rationally: Rational agent

- If you wanted to build an AI doctor, should it think exactly like a human doctor, or should it simply make the best medical decision?
- **Advantages** over other approaches
  - More **general** than the "laws of thought" approach. Laws of thought approach assumes that every decision can be made using logical rules. Can a self driving car only work using logic?
  - Google Maps uses traffic prediction, GPS, road conditions, and travel history. Is it only using logic?
  - More **obedient** to scientific development than approaches based on human behavior or human thought
  - **Standard of rationality (Logic)** is mathematically well defined and completely general
  - **Conclusion:**
  - Behave in right manner
  - Give max. Performance
  - Give Optimal Solution

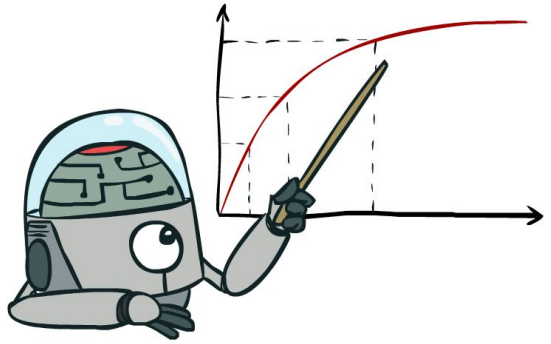
Designing Rational Agents



- An **agent** is an entity that *perceives* and *acts*.
- A **rational agent** selects actions that maximize its (expected) **utility**.
  - Rational: Choosing the best possible action given available information.
  - Utility: A numerical measure of how desirable an outcome is.
- Characteristics of the **percepts, environment,** and **action space** dictate techniques for selecting rational actions
  - **Self-Driving Car**
  - Percept: Camera sees red light
  - Agent: Decides to stop
  - Actuator: Apply brakes
- **This course** is about:
  - General AI techniques for a variety of problem types
  - Learning to recognize when and how a new problem can be solved with an existing technique

Rational Decisions

- We'll use the term rational in a very specific, technical way:
- Rational: maximally achieving pre-defined goals
- Rationality only concerns what decisions are made (not the thought process behind them)
- Goals are expressed in terms of the utility of outcomes
- Being rational means **maximizing your expected utility**



Approaches to AI

- **Strong AI**
  - Strong AI aims to build machines that can truly reason and solve problems who is self aware and whose overall intellectual ability is indistinguishable from that of a human being.
    - Human-Like
- **Weak AI**
  - Weak artificial intelligence (**weak AI**), also known as narrow AI, is artificial intelligence that is **focused on one narrow task**.
  - Deals with the creation of some form of computer-based artificial intelligence that **cannot** truly reason and solve problems, but can act as if it were intelligent.
  - **Siri** is a good example of narrow intelligence.

Weak and Strong AI

- **Weak AI** is AI that can not 'think', i.e. a computer chess playing AI does not think about its next move, it is based on the programming it was given, and its moves depend on the moves of the human opponent.
- **Strong AI** is the idea/concept that we will one day create AI that *can* 'think' i.e. be able to play a chess game that is not based on the moves of the human opponent or programming, but based on the AI's own 'thoughts' and feelings and such, which are all supposed to be exactly like a real humans thoughts and emotions and stuff.

Quick Quiz

- To what extent are the following computer systems instances of artificial intelligence:
  - Supermarket bar code scanners.
    - **Do they perceive the environment?** (they “read” the barcode)
    - **Do they act autonomously?** (only respond to human scanning)
    - **Do they adapt or learn?** (do not change behavior or improve)
    - **Do they have goals?** (no independent goal)
    - Conclusion: Not an AI agent, just a program executing fixed algorithm.
  - Web search engines.
  - Voice-activated telephone menus.

Applications of AI

- Game playing
- Computer vision
- Natural language processing
- Diagnosis systems
- Control
- Optimisation
- Robotics

Natural Language

- Speech technologies (e.g. Siri)
  - Automatic speech recognition (ASR)
  - Text-to-speech synthesis (TTS)
  - Dialog systems

- Language processing technologies
  - Question answering
  - Machine translation

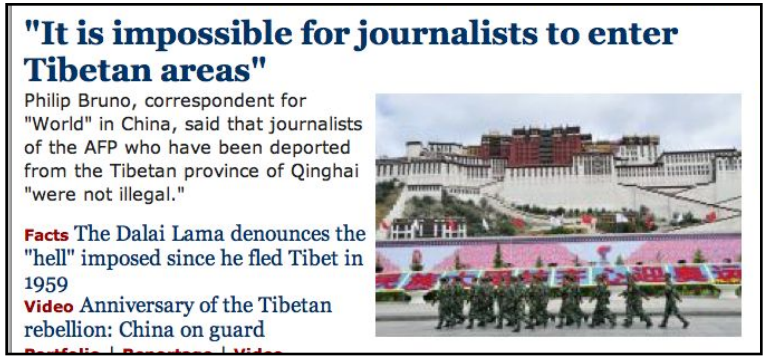


"Il est impossible aux journalistes de rentrer dans les régions tibétaines"

Bruno Philip, correspondant du "Monde" en Chine, estime que les journalistes de l'AFP qui ont été expulsés de la province tibétaine du Qinghai "n'étaient pas dans l'illégalité".

Les faits Le dalaï-lama dénonce l'"enfer" imposé au Tibet depuis sa fuite, en 1959

Vidéo Anniversaire de la rébellion



"It is impossible for journalists to enter Tibetan areas"

Philip Bruno, correspondent for "World" in China, said that journalists of the AFP who have been deported from the Tibetan province of Qinghai "were not illegal."

Facts The Dalai Lama denounces the "hell" imposed since he fled Tibet in 1959

Video Anniversary of the Tibetan rebellion: China on guard

  - Web search
  - Text classification, spam filtering, etc…

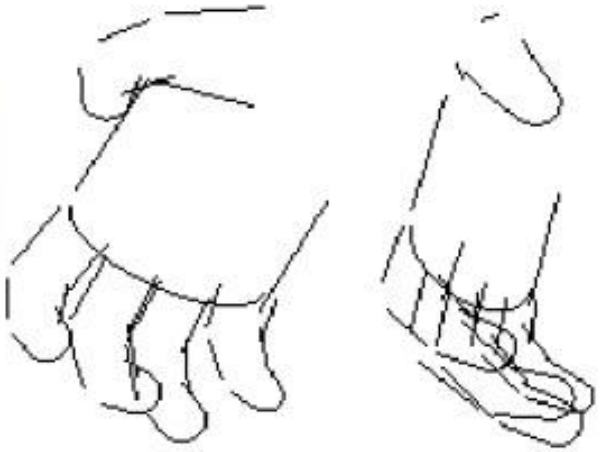
Vision (Perception)

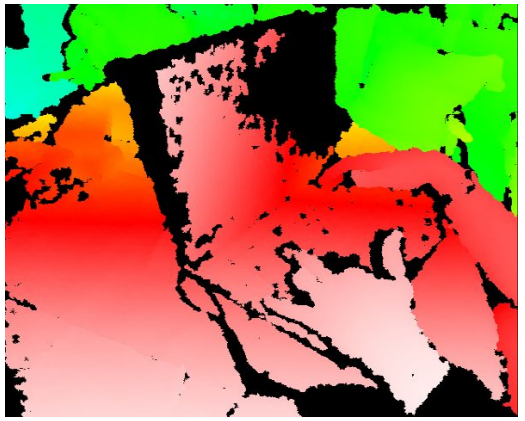
- Object and face recognition
- Scene segmentation
- Image classification (say hand pic as hand)











Robotics

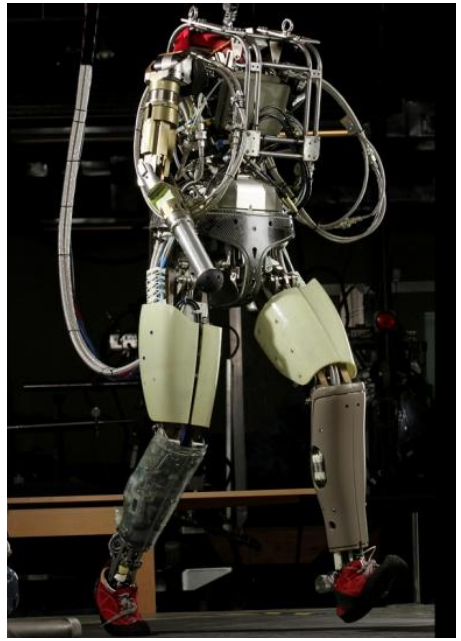
- Robotics

- Technologies
  - Vehicles
  - Rescue
  - Soccer!
  - Lots of automation…





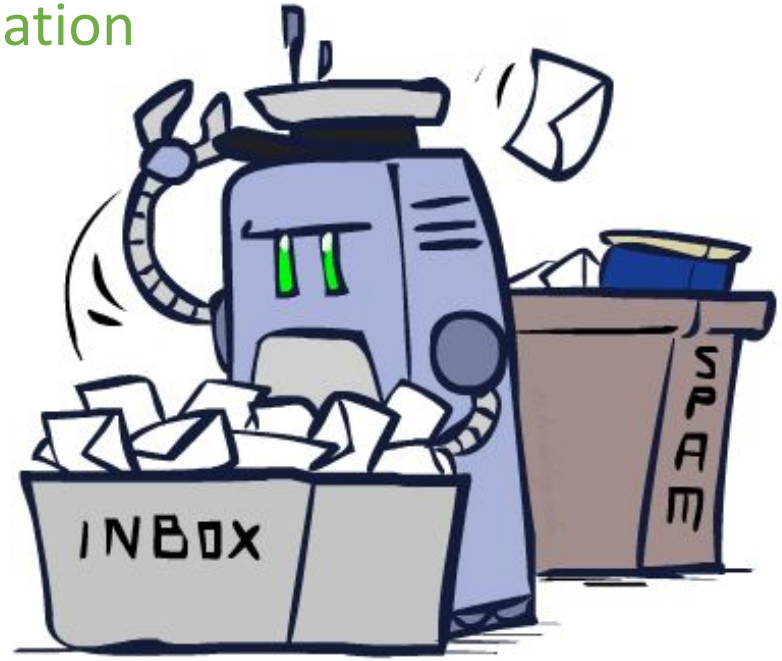




Decision Making

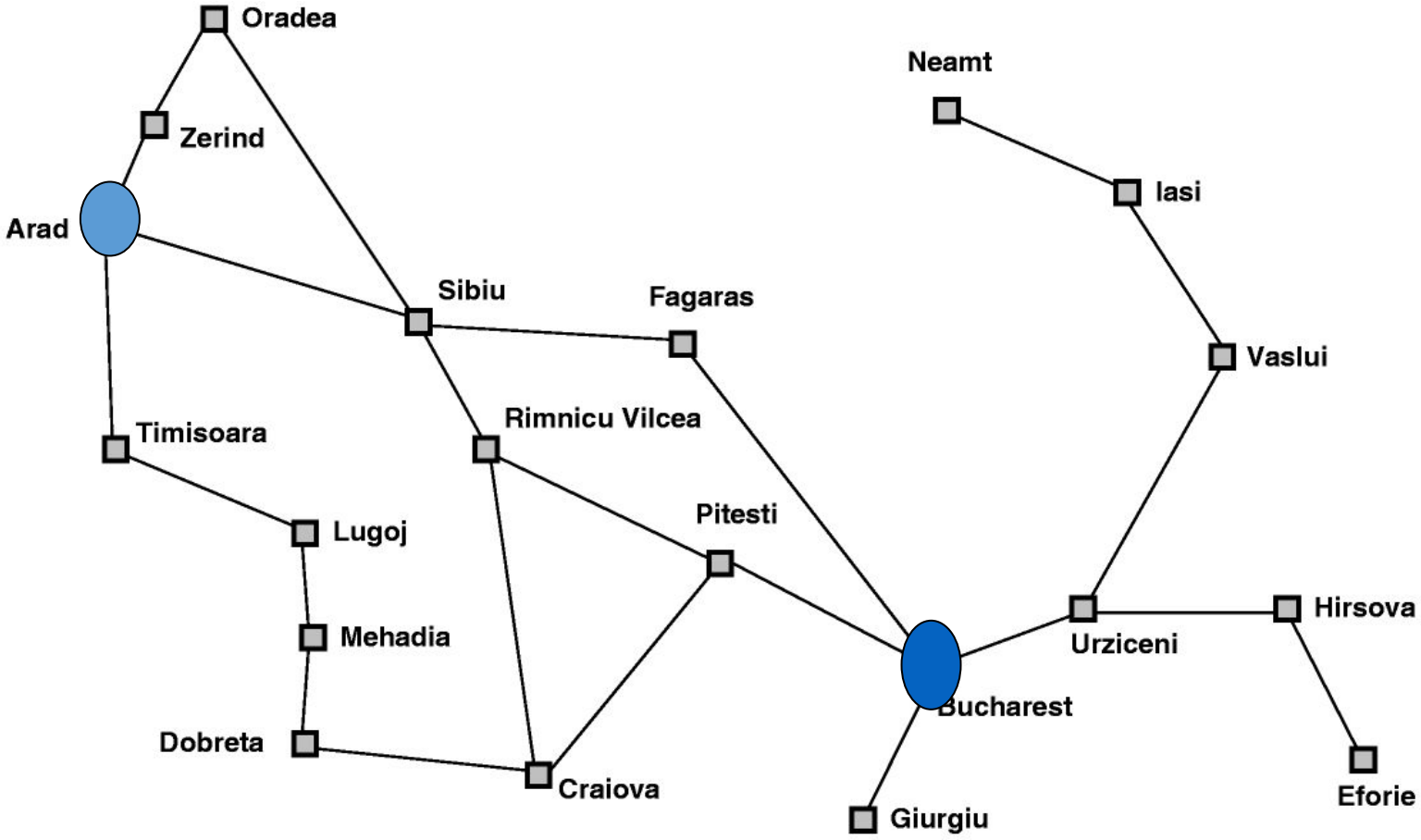


- Applied AI involves many kinds of automation
  - Scheduling, e.g. airline routing, military
  - Route planning, e.g. Google maps
  - Medical diagnosis
  - Web search engines
  - Spam classifiers
  - Automated help desks
  - Fraud detection
  - Product recommendations
  - … Lots more!



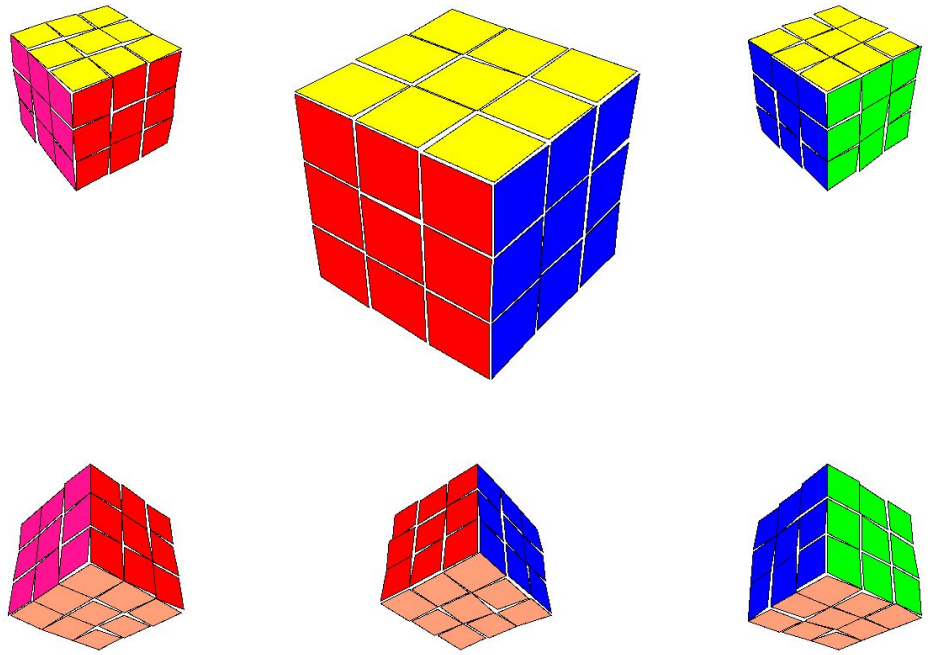
AI Topics: A Quick Introductory Overview

Classic AI search problems: Map searching (navigation)



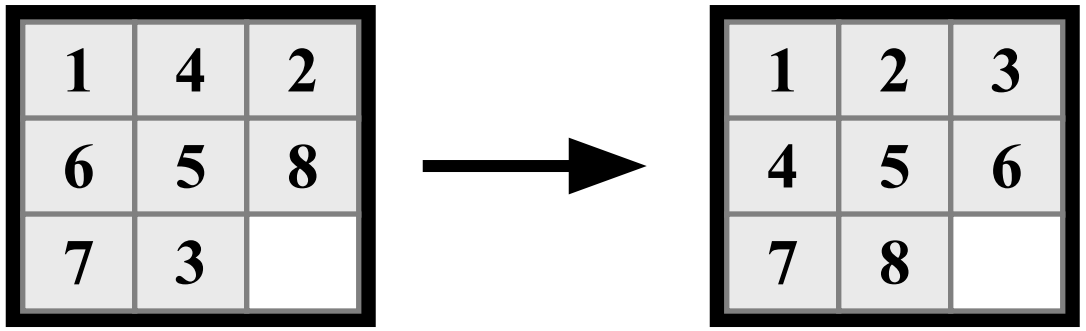
AI Topics: A Quick Introductory Overview

- Classic AI search problems
  - 3*3*3 Rubik's Cube



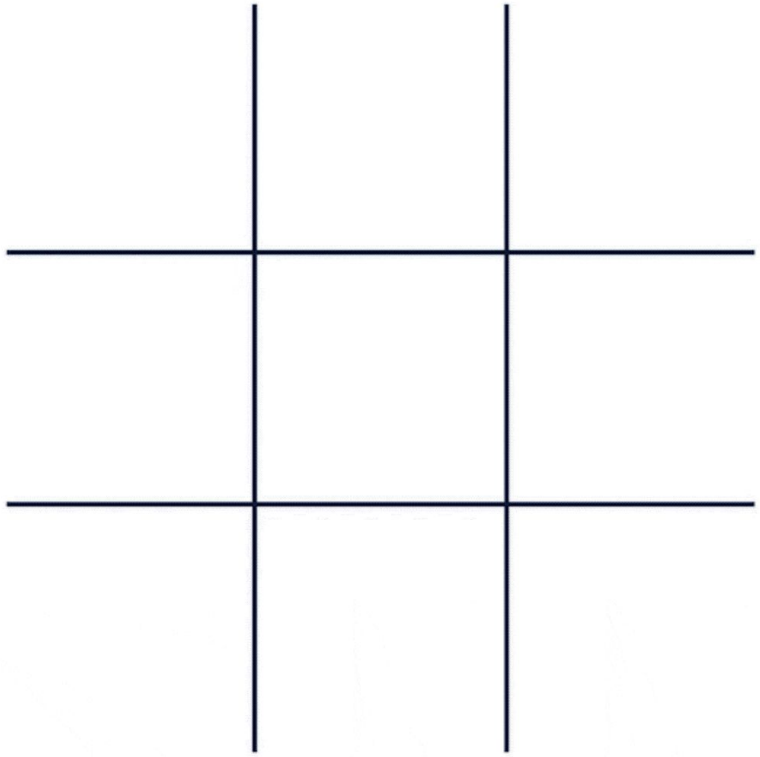
Classical AI Search Problems

- 8-Puzzle



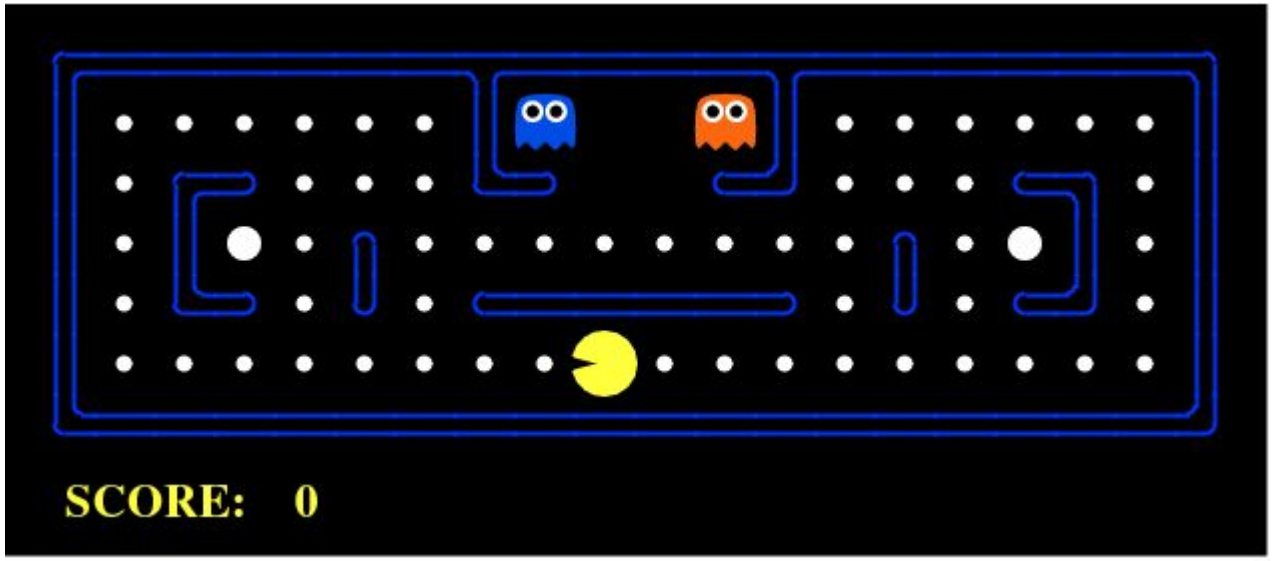
1	4	2
6	5	8
7	3	

Classical AI Search Problems



Classical AI Search Problem

- Pac-Man



Important Features of Artificial Intelligence

- The use of computers to do reasoning, pattern recognition, learning, or some other form of inference.

medical research papers to suggest cancer treatments based on symptoms and past cases.

- A focus on problems that do not respond to algorithmic solutions. This underlies the reliance on heuristic search as an AI problem-solving technique.

Instead of checking every possible move (which is impossible in real time), AI uses **heuristics** to evaluate the best moves.

- A concern with problem-solving using inexact, missing, or poorly defined information and the use of representational formalisms that enable the programmer to compensate for these problems.

Self-Driving Cars (Tesla, Waymo) – The AI must **navigate in foggy conditions, recognize road signs even if they are partially hidden**, and make real-time driving decisions with incomplete data.

- Reasoning about the significant qualitative features of a situation.

Medical Diagnosis AI – AI **doesn't just check symptoms but also considers the patient's history, lifestyle, and genetic factors** to recommend treatment.

- An attempt to deal with issues of semantic meaning as well as syntactic form.

Google Search – AI **understands the intent behind a user's search query**, not just the keywords. For example, searching "Apple" could mean **the fruit or the company**, and AI uses context to decide.

Important Features of Artificial Intelligence

- Answers that are neither exact nor optimal, but are in some sense “sufficient”. This is a result of the essential reliance on heuristic problem-solving methods in situations where optimal or exact results are either too expensive or not possible.

Spam Filters – AI **doesn’t perfectly classify every email**, but it **filters enough spam to make email useful** without blocking real messages.

- The use of large amounts of domain-specific knowledge in solving problems. This is the basis of expert systems.

AI was used in the **1970s to diagnose blood infections** based on an extensive database of medical knowledge.

- The use of meta-level knowledge to effect more sophisticated control of problem-solving strategies. Although this is a very difficult problem, addressed in relatively few current systems, it is emerging as an essential area of research.

AlphaZero (Game AI by DeepMind) – It **learns to play Chess by adjusting its own strategies dynamically**, improving over time **without human intervention**.

Self learning through trial and error, use reinforcement learning.